

# Evaluating the societal impact of disaster-driven infrastructure disruptions: A water analysis perspective

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## ABSTRACT

Critical infrastructure plays an important role in providing essential products and services to people; however, it is vulnerable to disasters and its disruption can have a severe societal impact. This study establishes a theoretical framework and proposes an integrated methodology to evaluate the societal impact of disrupted infrastructure in disaster events. Here, the societal impact is based on a water analysis and is defined as the percentage of the population in different levels of need satisfaction, which depends on the amount of water that people can obtain. Subsequently, based on water supply resilience analysis, an engineering model is proposed to estimate the water quantity during a disaster event considering the availability of tap water, emergency water, and bottled water. Meanwhile, physical interdependencies among the water supply system, emergency service, commercial facilities, electricity system, and transportation system are incorporated into an availability analysis of the three water sources. Finally, the proposed model is applied to the city of Osaka to validate its applicability and effectiveness. The results demonstrate that the proposed model can provide valuable information to decision-makers about the severity and spatial distribution of the societal impact of infrastructure disruption.

## 1. Introduction

Modern societies comprise several levels of interdependent critical infrastructure (CI), which provide people with essential resources and services, e.g. water, food, housing, and medical services [1]. Infrastructure can be disrupted due to damage caused when natural and man-made hazards strike a modern city. In such situations, residents would have difficulties in accessing life-supporting resources and some people's basic needs may not be satisfied, which could lead to severe societal impacts. As cities continue to develop, residents are becoming increasingly dependent on CI, and the societal impact due to unexpected disturbances become larger. Therefore, evaluating the negative impact of infrastructure disruption on society is critical for providing essential information to decision makers and to ensure minimal impact upon the daily lives of residents in the aftermath of disasters.

There are several studies regarding the vulnerability [2,3], reliability [4,5], and resilience [6,7] analysis of infrastructure. Furthermore, the

interdependency among different infrastructure systems, which can lead to cascading failure propagations, has received increasing attention during the last decade [8–11]. Results of these studies have highlighted the vital role of CI in social and economic development. However, little effort has been made to connect various infrastructure disruptions with their societal impact, owing to the difficulties in quantitatively measuring the societal impact and integrating it with disruptions.

The societal impact of a disaster can be multifaceted due to the multidimensionality of society and can include socioeconomic, psychosocial, sociodemographic, and political impacts [12–14]. When focusing on the consequences caused by damaged CI, the impact can be measured using the well-being change dimension [15,16]. Tabandeh et al. [17,18] assessed the decrease in well-being due to a disaster via a reliability-based capability approach that considers the disruption of infrastructure. Deshmukh et al. [19] developed a framework to semi-quantitatively evaluate the changes in socio-economic activity due to disrupted infrastructure. However, many indicators used to measure

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impacts in these studies rely heavily on the experts' subjective judgement and may lead to estimation bias. Conversely, Dargin et al. [20] proposed a theoretical framework to assess the effect of lifeline service disruptions on the well-being of households. They used hardship experience as a proxy variable to measure the emotional aspects of well-being. Similarly, Esmalian et al. [21] and Dong et al. [22] utilised the hardship experience of individuals to measure the well-being impact and quantified it by comparing the service disruption severity and household tolerance levels. Overall, using hardship experience to measure changes to well-being is based on social investigation, so the results are region- and disaster-specific. Additionally, they did not examine the underlying reasons for negative emotional well-being, which is induced and escalated by the unsatisfactory fulfilment of residents' needs. Mahdavian et al. [23] illustrated that people's anticipated needs for water and food increase immediately after a blackout, and the unsatisfaction of essential service needs, especially information, could escalate the negative emotions of individuals. However, this mechanism has not been incorporated into the definition and evaluation of societal impacts in previous studies.

Incorporating needs satisfaction into the definition of societal impact can help connect infrastructure disruptions with societal impact evaluation. This involves evaluating the gap between the demand and supply of goods/services during a disaster event. Penchansky and Thomas [24] proposed five dimensions (accessibility, availability, affordability, accommodation, and acceptability) to evaluate the supply-demand disparity in the health care system. Nozhati et al. [9] proposed a probabilistic framework to predict failure of food supply, considering disrupted CI from three dimensions (accessibility, availability, and affordability of food) after a disaster. During a disaster event, the availability of necessary resources/services for individuals is mostly impaired due to functionality failure of multiple CI systems, and its reduction directly leads to the unsatisfaction of individuals' needs. Hence, estimating the availability of essential services after disasters is critical to predict their societal impact. This is one stream of infrastructure resilience analysis, in which the resilience is usually defined as the potential to remain operational under any disruption [25].

In the field of infrastructure resilience analysis, extensive research has been conducted with respect to service availability analysis during disruptions, such as medical, food, water, and housing services. Chang et al. and Yavari et al. [26,27] established a model to predict the functionality of medical services (hospitals) by integrating the performance of lifelines (electricity and water), building damage, and personnel. The availability of water supply systems is usually evaluated by network analysis and hydraulic simulation considering the failure of water system components and interdependent infrastructure [28–30]. For post-disaster studies, Nazarnia [25] incorporated the social network method to analyse water service disruption patterns using household service disruption data. The housing service of buildings, which is measured by population dislocation and outmigration, were estimated as a function of socio-economic factors, physical damage to buildings, electricity network condition, and water networks [26,31]. In general, available resources/services for individuals usually depends on the resilience of multiple infrastructure systems. Additionally, the adaptive capacity of CI is an important component of resilience, which influences the availability of life-supporting goods/services. For example, the capacity to supply substitute or emergency resources from institutional agencies in a disaster event [32–34]. However, few studies consider adaptive capacity when estimating the total amount of resources/services available to individuals, and this may overestimate the societal impact.

In view of the abovementioned challenges, this study proposes a theoretical framework and an integrated methodology to quantify the societal impact of infrastructure disruptions in the context of disasters. The main contributions of this study include the incorporation of individual needs satisfaction into the definition and evaluation of societal impact; and the estimation of availability of resources/services for

individuals considering the interdependency and adaptive capacity of infrastructure systems. The remainder of the study is organised as follows. In Section 2, a theoretical framework is proposed to illustrate how disrupted infrastructure can have a societal impact; we expand this framework from the perspective of a water analysis. In Section 3, an integrated methodology is presented to estimate societal impact considering the disruption of water-related CI systems, including water supply system, electricity system, transportation system, emergency service, and commercial facility. Then, Section 4 presents a case study of Osaka using the proposed methodology, and explains the corresponding results. Finally, a discussion of these results and the conclusions of the study are provided in Section 5.

## 2. Theoretical framework

The theoretical framework shown in Fig. 1 presents how infrastructure disruptions and societal impact are connected. In general, from the perspective of the individual, people consume essential goods/service, which are provided by CI, to perform necessary activities in order to satisfy their needs. According to Maslow's hierarchy of needs, the most fundamental needs are physiological needs, followed by safety and security, belonging, and growth and achievement needs [1,35]. In the context of disaster events, physiological needs, like food, water, housing, and clothing, are more urgent than others [1]. Thus, individual physiological needs are the focus of this study. If infrastructure systems are damaged due to a disaster, their service provision for individuals will be reduced. Consequently, people's daily activities may be affected, which can lead to people's needs not being satisfied. The different levels of satisfaction of everyone's needs can reflect the severity of the whole societal impact. Hence, this study defines societal impact as the percentage of a population whose needs cannot be satisfied due to the infrastructure disruptions caused by disasters.

To better illustrate the mechanism by which infrastructure disruptions can have a negative societal impact, this study focuses on water, as water plays a vital role in daily life. The definitions and categories of CI used in this study are adopted from the National Infrastructure Protection Plan (NIPP) [13,36]. As previously mentioned, the availability of resources/services depends on infrastructure resilience, which can be quantified as the functionality of systems providing services to customers following a disaster [1,37]. Meanwhile, infrastructure systems possess inherent resilience and adaptive capacity, that is, the ability or potential to respond successfully to a disruption (e.g. emergency response) [32–34]. Thus, it is necessary to incorporate adaptive capacity into hazard impact evaluation [38]. Considering the water supply resilience in the aftermath of disaster, there are three main water sources provided by CI: tap water (TW), bottled water (BW), and emergency water (EW), as shown in Fig. 1. Many studies have proved that the increased social capital of communities could also enhance the water supply resilience; however, this was not considered in our study [39,40]. TW is the main water source provided by water supply system, while the supply of BW and EW reflects the adaptive capacity of the whole infrastructure systems. Specifically, BW is a TW substitute that residents can obtain from commercial facilities such as stores, malls, and supermarkets. The functionality of these commercial facilities influences the availability of BW. EW supply occurs in response to TW suspension, and mainly depends on the functionality of emergency services. Moreover, the electricity system and transportation system are considered in this study because the water supply of CI is physically dependent on them and, therefore, they influence the availability of water [33,41]. Simultaneously, water resource supports social functions of sustaining human activities, e.g. individual daily activities, production, and other activities [42,43]. Individual activities are emphasised in this study as they are achieved to satisfy human needs; for instance, drinking and cooking could satisfy the need for survival; washing, bathing, and laundry could meet people's basic hygiene needs. Whether people's needs are satisfied or not under disruptions determines the level of

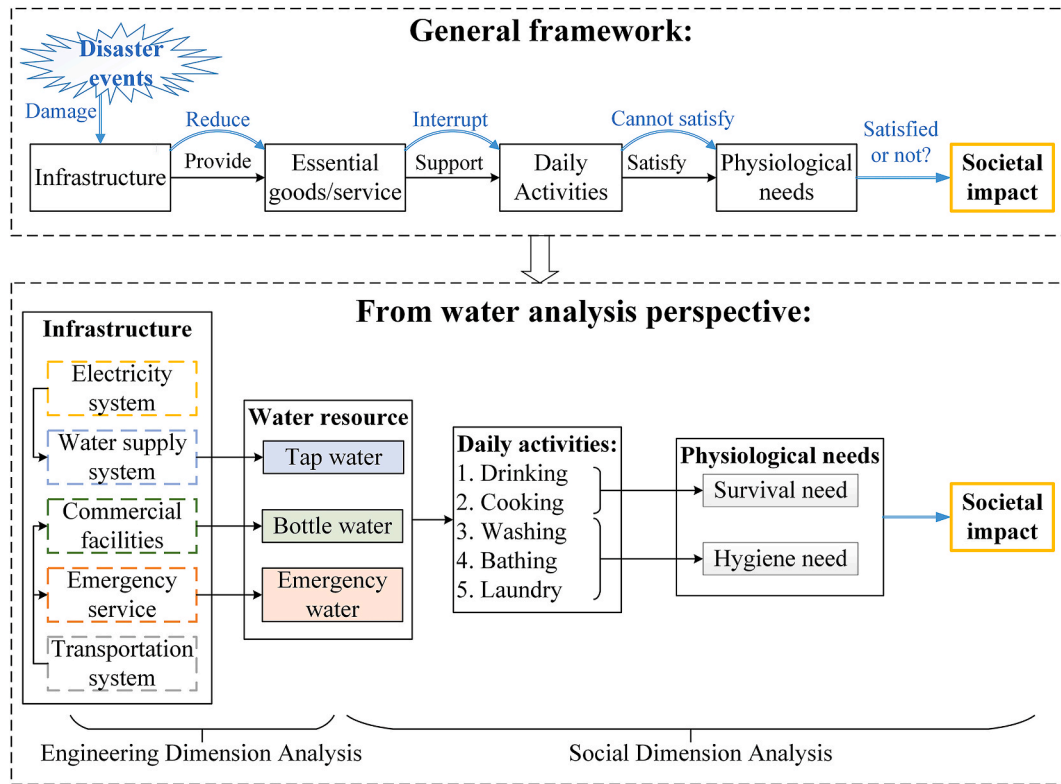


Fig. 1. Theoretical framework for societal impact analysis of the infrastructure disruption.

societal impact.

To quantitatively estimate the societal impact of infrastructure disruptions, it is critical to evaluate relationships among CI systems, essential resources/services, daily activities, and needs satisfaction. Therefore, we integrated a social dimension analysis and an engineering dimension analysis centred on the availability of essential services. The social dimension analysis involves evaluating the needs satisfaction level from the perspective of individuals, i.e., focusing on the societal adverse effect estimation. The engineering dimension analysis is related to the resilience analysis of infrastructure; the availability of essential services is estimated considering disruptions to CI systems.

### 3. Methodology

The following notation is used in the following sections:

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| $WQ$           | Water quantity available to individuals                                                                  |
| $avail_i^{TW}$ | Availability of TW from demand node $i$                                                                  |
| $avail_j^{BW}$ | Availability of BW from commercial facility $j$ (stores, markets, etc.)                                  |
| $avail_k^{EW}$ | Availability of EW from shelter $k$                                                                      |
| $pressure_i$   | Water pressure at demand node $i$                                                                        |
| $CON_{jd}$     | Connectivity between commercial facility $j$ and distribution centre $d$                                 |
| $ELE_j$        | Availability of electricity in commercial facility $j$                                                   |
| $PHS_j$        | Physical state of commercial facility $j$                                                                |
| $TW_k$         | Total water quantity in shelter $k$                                                                      |
| $POP_k$        | Number of residents served by shelter $k$                                                                |
| $INV_k$        | Water inventory in shelter $k$                                                                           |
| $V_n$          | Water volume capacity of the truck $n$                                                                   |
| $x_{kn}$       | Number of visits of truck $n$ to shelter $k$ . Each truck trip is assumed to visit one shelter at a time |
| $T_{kn}$       | Travel time from the location of the truck $n$ (water supply sources) to the shelter $k$                 |
| $DT$           | Waiting time for loading and unloading water in one trip                                                 |
| $SD_{kn}$      | Shortest distance from location of the truck $n$ to the shelter $k$                                      |
| $Speed_n$      | Travel speed of the water truck $n$                                                                      |
| $PC$           | Purification capacity of the EW supply sources                                                           |

Based on the theoretical framework presented in Section 2, an integrated methodology was developed to quantify the societal impact of disrupted infrastructure. From the social dimension analysis, a quantitative relationship between the societal impact estimation and quantity of available water was established. From the engineering dimension analysis, an integrated water quantity estimation model was developed to estimate the availability of water in a disaster event considering the disruption to and interdependency of CI systems (i.e. the electricity system, the water supply system, the transportation system, emergency services, and commercial facilities). By connecting the engineering and social dimension analyses using water quantity, the societal impact can be estimated given direct damages to infrastructure systems during a disaster event.

#### 3.1. Societal impact estimation of water disruption

Societal impact, here, is defined as the percentage of a population whose physiological needs cannot be satisfied due to disrupted infrastructure. In other words, the level of needs satisfaction for every individual contributes to the societal impact as a whole. Whether needs are satisfied or not depends on the achievement of the corresponding activities for each individual. Furthermore, the achievement of activities relies on the provision of essential goods/service by CI.

Water is fundamental to maintain essential daily activities, such as drinking, cooking, washing, bathing, and laundry. The water supply in the society, which is illustrated in Fig. 1, is susceptible to disasters, for example, earthquakes, typhoons, and storm surges [39]. This could reduce the provision of TW, BW, and EW, restricting people's abilities to conduct daily activities. As listed in Table 1, five levels of needs satisfaction are defined according to the achievement of different activities from the perspective of individuals [44]. According to the characteristics of people's daily activities, the achievement of the activities is a function of the quantity and quality of water [43]. However, considering poor-quality water, which is not meeting water quality standards, will not be supplied to citizens in a disaster event [45], we assume that the

**Table 1**

Relationship among needs-satisfaction levels, activities, and water quantity (adopted from Ref. [44]).

| Levels  | Satisfaction of needs                                 | Achievements of daily activities                                                                | Water quantity (unit: L/d) |
|---------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------|
| Level 1 | Survival needs can be met; Hygiene needs can be met   | Drinking, cooking, washing, bathing, and laundry can be assured                                 | $WQ \geq 100$              |
| Level 2 | Survival can be met; Hygiene can mostly be met        | Drinking, cooking, and washing can be assured; Bathing and laundry is possible                  | $20 \leq WQ < 100$         |
| Level 3 | Survival can be met; Hygiene can partly be met        | Drinking and cooking can be assured; Washing is possible; Bathing and laundry cannot be assured | $5 \leq WQ < 20$           |
| Level 4 | Survival can mostly be met; Hygiene can partly be met | Drinking can be assured; Cooking is possible; Others cannot be assured                          | $2 \leq WQ < 5$            |
| Level 5 | Survival needs and hygiene needs cannot be met        | No activity can be assured                                                                      | $0 \leq WQ < 2$            |

achievement of people's activities is a function of the quantity of water.

The water required for individuals' activities varies with climatic conditions, lifestyle, culture, wealth, and other factors. Many studies have investigated and recommended the minimum water requirement of activities for most people under most conditions, which can provide valuable information in emergency situations [43,44,46]. For example, Gleick recommended that the minimum water requirements for drinking, sanitation, bathing, and cooking are 5, 20, 15, and 10 L per day, respectively [43]. The minimum standards in the Sphere Handbook for individuals' drinking, hygiene, and cooking are 2.5–3, 2–6, and 3–6 L per day, respectively [46]. This study adopted the results from the work of Howard and Bartram because their work also included the allocation of water for each activity with limited water quantity (Table 1) [44]. Given a certain amount of water, the water used for drinking and cooking are generally prioritised to meet people's survival needs. With an additional volume of water, people can consider personal hygiene, washing, bathing, and laundry activities. Different achievements of

activities and the corresponding water quantity are shown in Table 1. Moreover, five levels of needs satisfaction, which are defined according to the achievement of daily activities, can be linked to the quantity of water (Table 1). Thus, based on this quantitative relationship, the needs satisfaction level of each individual can be evaluated given the different spatial distribution of water quantity due to damaged CI. Then, the percentage of the population at different levels can be derived to reflect the quantitative societal impact.

### 3.2. Integrated water quantity estimation model

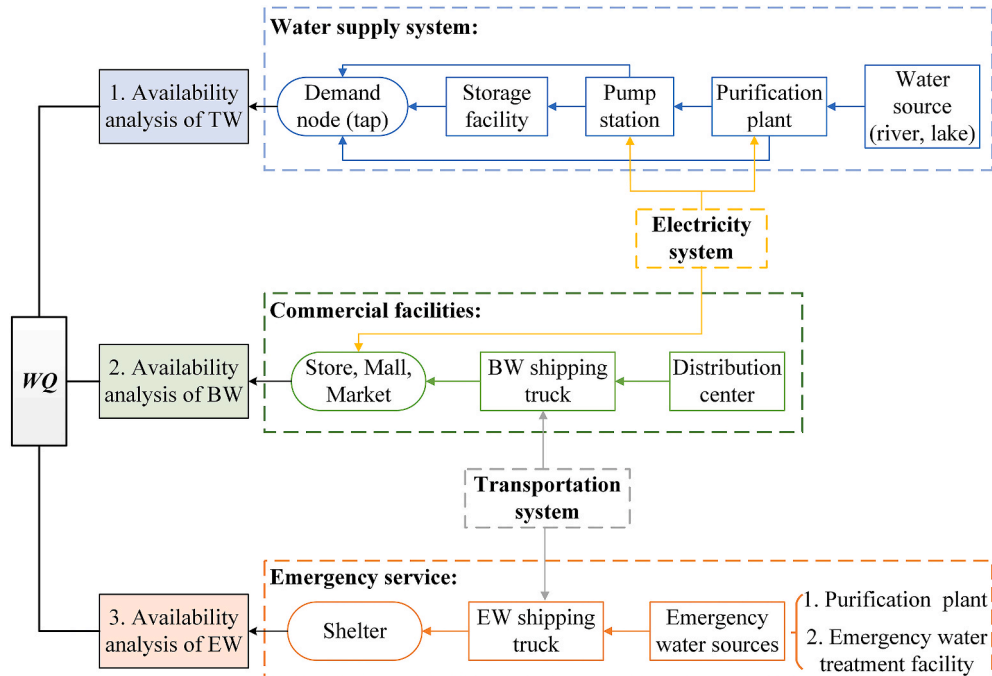
This section proposes an integrated model to predict the available water quantity for residents during a disaster event. Based on the theoretical framework in Section 2, the water quantity estimation model is composed of an availability analysis of TW, BW, and EW (Eq. (1)). These three water sources are mainly provided by the water supply system, commercial facilities, and emergency services, respectively. Meanwhile, owing to the physical interdependency among infrastructure systems, the transportation and electricity systems are also incorporated into the water estimation model. The interdependencies among CI systems are detailed in Fig. 2. The mathematical function of the integrated water quantity estimation model is as follows:

$$WQ = avail_i^{TW} \cup avail_j^{BW} \cup avail_k^{EW} \quad (1)$$

According to the characteristics of TW, BW, and EW, this study made the following assumptions:

- If TW from the demand node  $i$  in the water supply network is available ( $avail_i^{TW} = 1$ ), individuals can receive a normal quantity of water ( $WQ = 100\text{L/d}$ ), and EW will not be delivered to the residents who can access TW.
- BW can only support the consumption activities of people, including drinking and cooking. The water quantity from BW is restricted to 3 L/d for individuals if available ( $avail_j^{BW} = 1$ ).
- The actual quantity of EW that each individual can obtain in shelter  $k$  is  $avail_k^{EW}$ , and EW can be used for all activities.

The detailed methods to estimate the available amount of TW, BW,



**Fig. 2.** Water availability analysis and relationships among CI systems.

and EW in a disaster scenario are introduced in Sections 3.2.1–3.2.3.

### 3.2.1. Availability analysis of tap water

The water supply system, which provides TW for residents, is the main source of water. It purifies water from a river, lake, or well in a purification plant, and transports the treated water to every household through water pipelines to ensure that people have access to clean water from household taps. As indicated in Fig. 2, the essential components of the water supply system are the purification plant, pump station, storage node, and water pipeline. Pump stations play an important role in ensuring the delivery of clean water by providing enough pressure. Storage facilities, like water tanks, water towers, and water reservoirs, are used for storing treated water and ensuring a steady water supply when the water demand is high. Pump stations and purification plants cannot operate without electricity, which is mainly provided by the electricity system. Standby power systems are used in case of a power outage in the aftermath of a disaster.

After a disaster, some components of the water network can be directly or indirectly damaged, which could lead to the unavailability of water at demand nodes (the aggregated water demand of households in a service area). For example, more than one million people lost TW following Hurricane Katrina in the southern United States, and almost 100% of people lost water services after the 2010 Chile earthquake [39]. A connectivity analysis of water supply systems was performed to evaluate whether treated water can be delivered from the source to demand nodes or not after the failed components were removed [28]. However, the connectivity analysis could not adequately model the availability of water from taps because water cannot be transported to taps without sufficient pressure and flow in the network; thus, a hydraulic analysis is also needed [47]. There are many hydraulic simulation tools for water network systems, amongst which EPANET is the most popular [7,29,48,49]. However, EPANET, which uses a demand-driven hydraulic engine, is not suitable for pressure-deficient situations that often occur in disaster events [27]. In order to solve this problem, Pathirana developed the EPANET-EMITTER package using a pressure-driven approach on the basis of the EPANET solver [50]. This package is suitable for low pressure analyses and has a modest input data requirement. Therefore, the EPANET-EMITTER package was adopted in this study to build a water supply system and performed a hydraulic simulation.

The availability of TW is related to the network topology, pressure, and flow. Pressure is a more important factor to consider than topology or flow when measuring the availability of TW, because the pressure value can reflect the connectivity and flows of system. A pressure of zero indicates no connectivity between the purification plant and the demand node. In addition, the flow rate at the demand node is a function of pressure and is reduced when the pressure is below a certain threshold, which is set to 10 m based on the literature [50]. Moreover, low pressure increases the possibility of the potable water becoming contaminated in the water network. Some regions allow non-potable water (low pressure nodes) to be transported to customers while ensuring that customers are informed of methods to treat the water, whereas others choose to terminate the supply of water with poor quality [45]. This study adopts the latter strategy, in which poor quality water is not delivered to households, and assumes that water has a poor quality at demand node  $i$  if the pressure falls below a minimum value (10.6–14.1 m) [47,48]. Therefore, the availability of TW at every demand node  $i$  is a function of pressure as follows:

$$avail_i^{TW} = \begin{cases} 0 & \text{pressure}_i \leq 10.6\text{m} \\ 1 & \text{pressure}_i > 10.6\text{m} \end{cases} \quad (2)$$

From the hydraulic simulation in the EPANET-EMITTER software, and considering the disrupted components of the water supply system (damages mapping algorithms are shown in Table 2), the updated pressure at the demand nodes can be calculated. Then, according to the

**Table 2**

Mapping algorithm for damaged components of the water supply system in EPANET-EMITTER (adapted from Ref. [47,51]).

|   | Damaged components                                     | Mapping approach in EPANET-EMITTER                                                                                                          |
|---|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Pipeline                                               | Delete corresponding links between nodes                                                                                                    |
| 2 | Pump                                                   | Replace pump links with linear elements                                                                                                     |
| 3 | Water storage facility (water tank, water tower, etc.) | Remove the damaged storage facilities in the network and connect the water storage facility's inflow pipes with corresponding outflow pipes |
| 4 | Purification plant                                     | Set the outflow of the purification plant to zero                                                                                           |
| 5 | Electricity                                            | Replace supporting pump links with linear elements                                                                                          |

availability function of the water supply system (Eq. (2)), the demand nodes without TW can be identified.

### 3.2.2. Availability analysis of bottled water

In the aftermath of a disaster, BW provided by commercial facilities is another choice available to residents, and Beatty et al. [52] shown the evidence that the sales for BW increase sharply in commercial facilities after hurricane landfalls. As presented in Fig. 2, BW is transported from the distribution centre to stores, malls, markets, etc. by trucks, and these logistics heavily depend on the condition of the transportation system. The availability of stores, malls, or markets also depends on the electricity system because they cannot function without light, refrigeration, or Point of Sale (POS) machines. It is worth noting that commercial facilities are vulnerable to disasters and their physical damages could lead to the suspension of services as well. Thus, the availability of BW facilities can be calculated as:

$$avail_j^{BW} = \left( \sum_{d=1}^D CON_{jd} \right) \cap ELE_j \cap PHS_j \quad (3)$$

where  $avail_j^{BW}$ ,  $CON_{jd}$ ,  $ELE_j$ , and  $PHS_j \in \{0, 1\}$ .  $CON_{jd}$  denotes the connectivity between commercial facility  $j$  and distribution centres  $d \in \{0, 1, 2, \dots, D\}$ ,  $ELE_j$  represents the availability of electricity in facility  $j$ , and  $PHS_j$  signifies whether facility  $j$  is directly damaged (equal to 0 if the facility is damaged, while equal to 1 if the facility is not damaged). Although not all influencing factors were covered in this study, our model maintained an adequate amount of detail to study the availability of BW in commercial facilities.

### 3.2.3. Availability analysis of emergency water

Governments or water companies would take measures to ensure EW supply in the aftermath of a disaster if TW is unavailable. For instance, in 2015 Nepalese earthquake and other disasters, EW supply is proved to be an essential water source for residents [33,45]. The primary approach is using water trucks to deliver the treated water from EW sources (Fig. 2) to temporary water tanks in shelters, such as wide-area shelters, accommodation shelters, and nearby city parks. Thus, the amount of water that can be provided for people in emergency shelters is a function of the emergency strategies, transportation system, water purification capacity of EW sources, and the number of water trucks. In order to estimate the maximum amount of EW that an individual can access in a shelter, we adopt an equal water distribution strategy. This is because the emergency response strategy that governments usually choose is to provide equal amounts of water to every individual considering the limitation of resources [53,54].

This study proposes the following integer linear optimization model to estimate the maximum EW ( $avail_k^{EW}$ ) that each person can access in each shelter  $k$  based on the equal distribution strategy and limited resources:

$$\text{Max} \left\{ \min_{k \in \Omega} \text{avail}_k^{\text{EW}} = \frac{TW_k}{POP_k} \right\} \quad k = 1, 2, 3 \dots K \quad (4)$$

$$TW_k = INV_k + V_n * \sum_{n=1}^N x_{kn} \quad \begin{matrix} k = 1, 2, 3 \dots K \\ n = 1, 2, 3 \dots N \end{matrix} \quad (5)$$

subject to

$$\sum_{k=1}^K x_{kn} * (2 * T_{kn} + DT) \leq 24 \quad \begin{matrix} k = 1, 2, 3 \dots K \\ n = 1, 2, 3 \dots N \end{matrix} \quad (6)$$

$$T_{kn} = \frac{SD_{kn}}{\text{Speed}_n} \quad (7)$$

$$\sum_{n=1}^N \sum_{k=1}^K x_{kn} * V_n \leq PC \quad \begin{matrix} k = 1, 2, 3 \dots K \\ n = 1, 2, 3 \dots N \end{matrix} \quad (8)$$

$$x_{kn} \geq 0, x_{kn} \in \text{integer} \quad (9)$$

The objective function (4) maximises the minimum value of the water quantity per person in all shelters that belong to region  $\Omega$ , and its optimal value must be such that the water will be evenly distributed to each shelter [53]. As stated in Eq. (5),  $TW_k$  consists of the original inventory water  $INV_k$  and the water delivered by all trucks that choose shelter  $k$ . The decision variable is  $x_{kn}$ , which denotes the times that the truck  $n$  visits the shelter  $k$ . The constraints in Eq. (6) indicate the work time restriction (within 24 h) associated with each water truck  $n$ , where  $2 * T_{kn}$  represents the back and forth travel time from the location of the truck  $n$  to the shelter  $k$ , and  $DT$  denotes the loading and unloading time for one trip. The travel time of each water truck,  $T_{kn}$ , is derived from dividing the shortest path ( $SD_{kn}$ ) by the truck speed (Eq. (7)). Additionally, the shortest path  $SD_{kn}$  will be re-planned and re-calculated using network analysis in ArcGIS if disruptions to the transportation network occur [55]. The purification capacity  $PC$  of the EW supply sources is shown to have constraints (Eq. (8)). Finally, the constraints in Eq. (9) indicate that decision variables  $x_{ij}$  are non-negative integers.

To solve this max-min problem, the objective function in Eq. (4) can be alternatively formulated by maximizing an additional variable  $\lambda$  which is a lower bound for EW quantity per person in each shelter. Thus, Eq. (4) can be rewritten as Eqs. (10)–(11) in the distribution model, while the other original constraints remain unchanged in Eqs. (5)–(9):

$$\text{Max } \lambda \quad (10)$$

subject to

$$\frac{TW_k}{POP_k} \geq \lambda \quad k = 1, 2, 3 \dots K \quad (11)$$

Eqs. (5)–(9)

Eqs. (5)–(11) constitute the EW equal distribution model, which would determine the quantity of EW that each individual can access in the shelter  $k$  within one day. Indeed, this distribution model is an integer linear programming model, and this study utilises the *GUROBI solver* in Python to yield an optimal solution of this model [56].

#### 4. Osaka case study

In this section, the proposed model is used to analyse the societal impact of infrastructure disruption based on real-world and inferred data of the city of Osaka. Osaka is one of the most populous cities in Japan, with a total estimated population of 2.69 million. As mapped in Fig. 3, there are 24 districts in Osaka and the size of the population grid is  $500 \times 500$  m (analysis units). In addition, Osaka is threatened by various natural hazards, such as the earthquakes, floods, typhoons, and storm surges, which can lead to the failure of CI and affect residents' daily lives. This study conducts a scenario analysis to address the following two questions: (1) How should the proposed model be applied to evaluate the societal impact due to post-disaster infrastructure disruption in Osaka? (2) How does societal impact change due to different disruption severities?

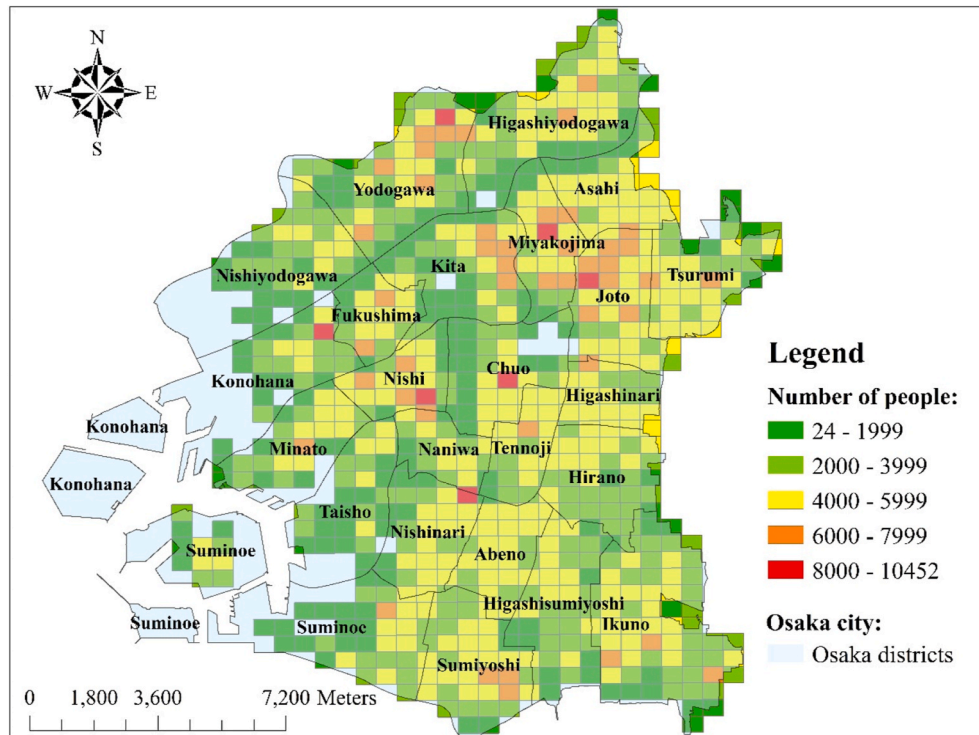


Fig. 3. Twenty-four districts of Osaka and their corresponding population distribution.

#### 4.1. Data description

The Osaka water supply system is operated by the Osaka Municipal Waterworks Bureau (OMWB), and its main components are presented in Fig. 4. Three water purification plants, with a total purification capacity of 2,430,000 m<sup>3</sup>/d, provide the TW for the whole city. The TW is distributed by 12 distribution plants (including the storage reservoir and pump station) to ensure the stability of the water supply (Fig. 4 (a)). The total length of the distribution pipelines is approximately 5200 km. There are 611 demand nodes, in which the base demands are estimated by multiplying the number of people in the service area by hourly average water consumption per person (Fig. 4 (a)). The transportation network data and commercial facility data (e.g. convenience stores, supermarkets, and malls) of Osaka indicated in Fig. 4 (b) were derived from OpenStreetMap [57]. The population grids, emergency shelters, and distribution centres were obtained from Japanese National Land Numerical Information [58].

#### 4.2. Illustrative case analysis

To quantitatively estimate the societal consequence caused by infrastructure failures, this study assumes the following disaster scenario: a strong typhoon and its induced storm surge struck Osaka in Japan. Kunijima purification plant was disrupted due to contamination by the storm surge, and the electricity in Higashiyodogawa district was interrupted, which lead to the malfunction of the pumps in the Kunijima distribution plant. Additionally, this study assumes that 5% of roads were disrupted randomly, which means that 5% of road segments (7588 road segments) were selected randomly from all roads (151,769 segments in total) and closed. Similarly, 1% of commercial facilities were selected randomly to suffer from direct damage. These disruptions were applied to the proposed methodology to evaluate the societal impact within Osaka.

##### 4.2.1. Tap water estimation

The water supply network for the city of Osaka was established using the EPANET-EMITTER software, and the pressure at each of the demand nodes was simulated under normal conditions. The simulation results are presented in Fig. 4 (a) and, to some extent, they can be validated by Javanbarg's research [59]. Then, considering the disruption scenario, which is the failure of the Kunijima purification plant and disruption of the pump station, the structure of the water network was updated according to the mapping algorithm shown in Table 2. Then, we reconducted the hydraulic simulation and updated the corresponding water pressure in each demand node. Finally, the availability of TW in each demand node and in each service area was evaluated according to Eq. (2). As indicated in Fig. 5, the red areas are the regions without TW under the assumed disaster scenario settings; however, these results cannot be verified because Osaka has not experienced the interruption of TW supply due to disaster.

##### 4.2.2. Bottled water estimation

In the analysis of BW, only commercial facilities in areas without TW were evaluated (Fig. 6). We used ArcGIS to build the transportation network of Osaka and then perform an Origin-Destination (OD) cost analysis to calculate the connectivity from distribution centres to stores, considering the failure of transportation (5%) [55]. It is worth noting that the disruption of transportation was modelled by generating barrier points in randomly selected road or bridge segments. Combined with the availability of electricity and direct damage (1%) of commercial facilities, the availability of BW in every facility was calculated using Eq. (3). Commercial facilities do not have certain serviceable areas that cover the whole region; hence, we assumed that if there was an available store in the population grid, the residents in that grid could access BW. Finally, the population grids with available BW were identified, and these are shown as green grids in Fig. 6.

##### 4.2.3. Emergency water estimation

As for areas without TW, the OMWB will deliver EW from the Niwakubo purification plant to the corresponding shelters, such as parks and accommodation shelters, via 21 water trucks (shown in Fig. 7). Similar to the connectivity analysis in the BW model, the travel costs from purification plants and shelters were calculated using an OD cost analysis based on the Osaka transportation network with 5% disruptions. Other important details were as follows: the volume of each truck was 4 m<sup>3</sup>; the speed of the trucks was assumed to be 40 km/h; the purification capacity of Niwakubo for EW was 128,000 m<sup>3</sup>, which is the total capacity minus the water supplied in the water network; the population served by each shelter was estimated by summing the number of people in its service area that were assigned by the proximity principle; the water inventory for the park was 0; and the water inventory in accommodation shelter was estimated by multiplying 0.2, the capacity of the shelter, and 3 L/d. Then, these parameters were applied to the emergency equal distribution model to estimate the maximum water quantity that every individual could access in shelters ( $avail_k^{EW} = 5 \text{ L/d}$ ). Finally, the EW quantity (5 L/d) was mapped to each grid according to the service area of the shelter. The EW available areas are shown in Fig. 7.

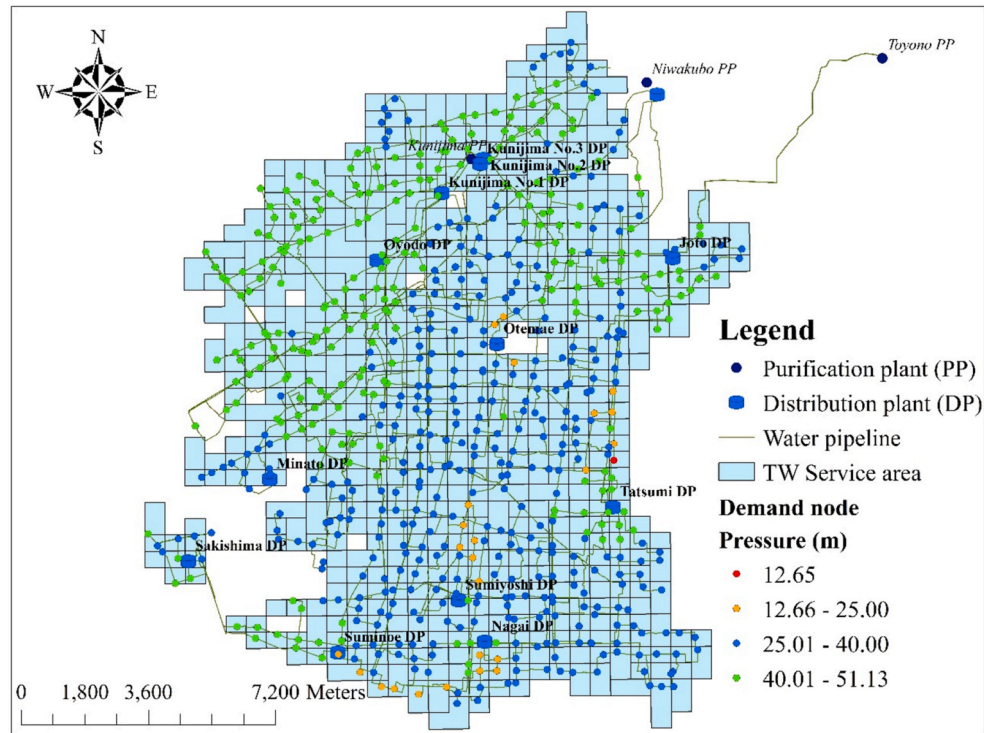
##### 4.2.4. Estimated societal impact of infrastructure disruptions

By integrating the availability of TW, BW, and EW, the proposed methodology can estimate the maximum water quantity that each individual can access in each grid (Eq. (1)). There were three possible water quantities under this disaster scenario (shown in Fig. 8), including 5 L (only EW was available), 8 L (EW and BW were available), and 100 L (TW is available). From the societal impact analysis, the people's needs-satisfaction levels in each grid can be derived according to the quantitative relationship in Table 1. The spatial distribution of the societal impact in Osaka due to infrastructure disruptions was estimated and mapped in Fig. 8. Overall, 84.6% of residents were in Level 1; all the needs of these people could be satisfied and daily water-related activities could be achieved. However, 15.4% of residents were in Level 3; only the survival needs of these people could be satisfied and some of their hygiene-related activities could not be achieved.

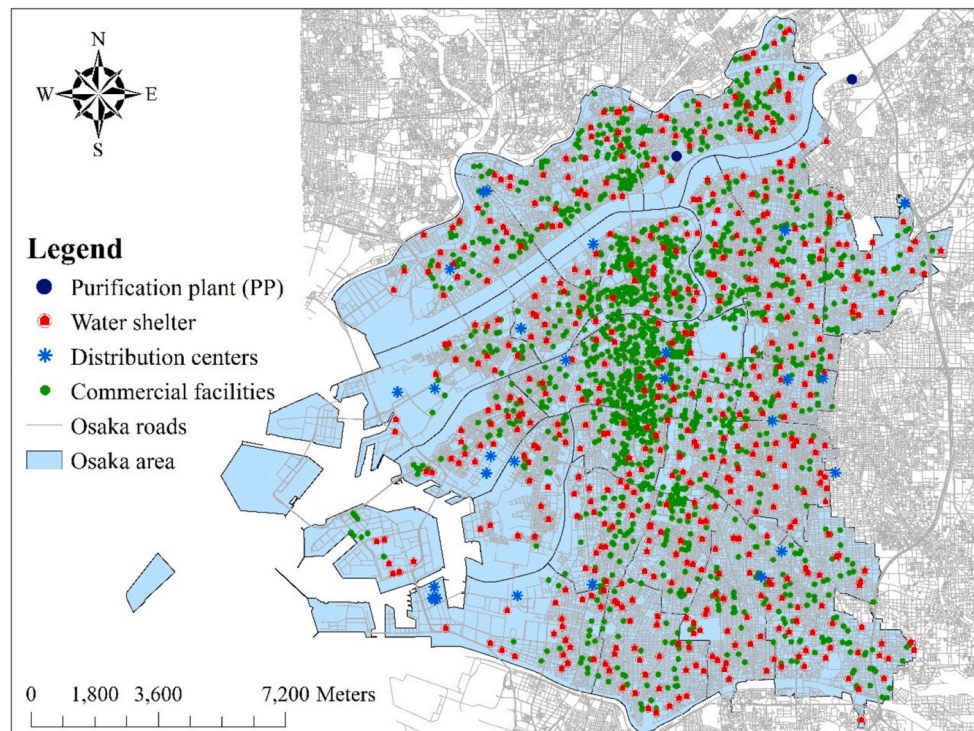
In general, as presented in Figs. 5 and 8, the societal impact pattern in Osaka showed the same characteristics with the spatial distribution of TW availability. This means that the impact was dominated by the availability of TW under this disaster scenario. Moreover, the availability of BW and EW, which are related to the adaptive capacity of infrastructure, were crucial in mitigating the negative societal impact in TW shortage areas because 15.4% of residents would have been exposed to Level 5 if BW and EW were not incorporated. Furthermore, by comparing the effects of EW and BW on societal impact (Figs. 6–8), it is found that the availability of EW was more important than BW in the regions without TW. This is because people's needs satisfaction could achieve Level 3 with the available EW (5 L/d) regardless of whether BW was integrated or not.

#### 4.3. Sensitivity analysis

This section analysed the effects of various infrastructure disruption scenarios on societal impacts. We focused on the electricity and transportation systems, as they are essential for supporting the supply of water. In mechanism, disruptions to the electricity system can lead to the malfunction of a pumping station in the water network system and cause the unavailability of TW. They can also disrupt commercial facilities and lead to unavailability of BW. Thus, analysing the disruption of electricity helps examine the effects of pumping station malfunctions and the unavailability of commercial facilities on societal impact. In addition, different damage ratios for transportation were simulated because the transportation system is essential for logistics, and transportation disruptions lead to inaccessibility and increased travel costs,



(a)



(b)

**Fig. 4.** Distribution of CI in Osaka: (a) water supply system under a normal situation, and (b) transportation system, commercial facilities, and emergency shelters.

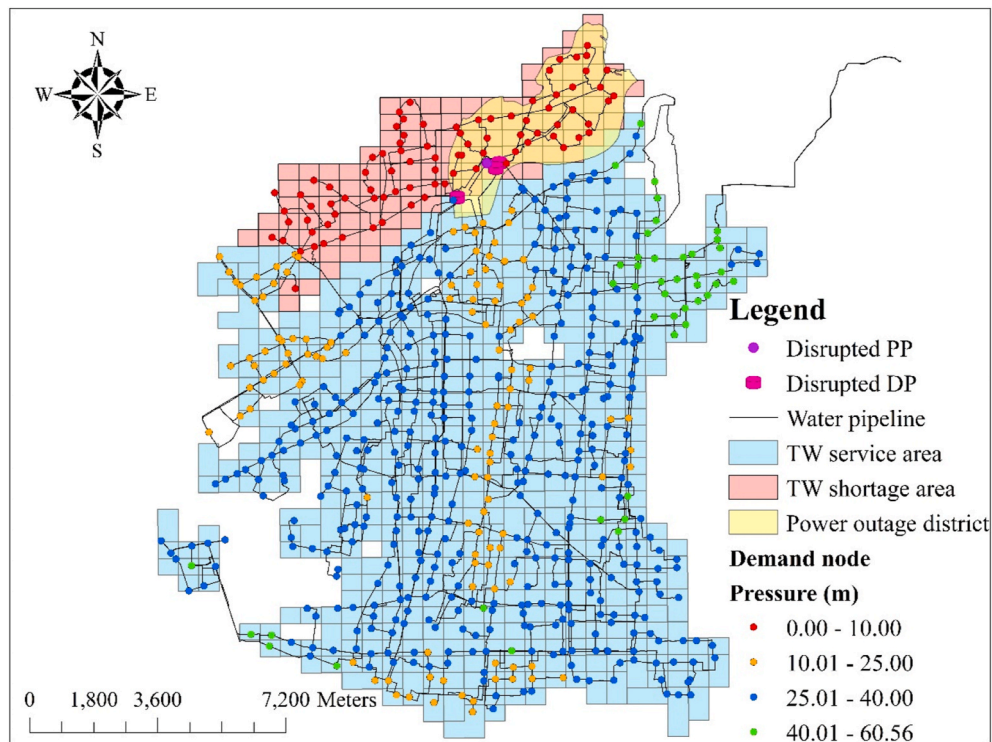


Fig. 5. Hydraulic simulation results of the water supply system under the disaster scenario.

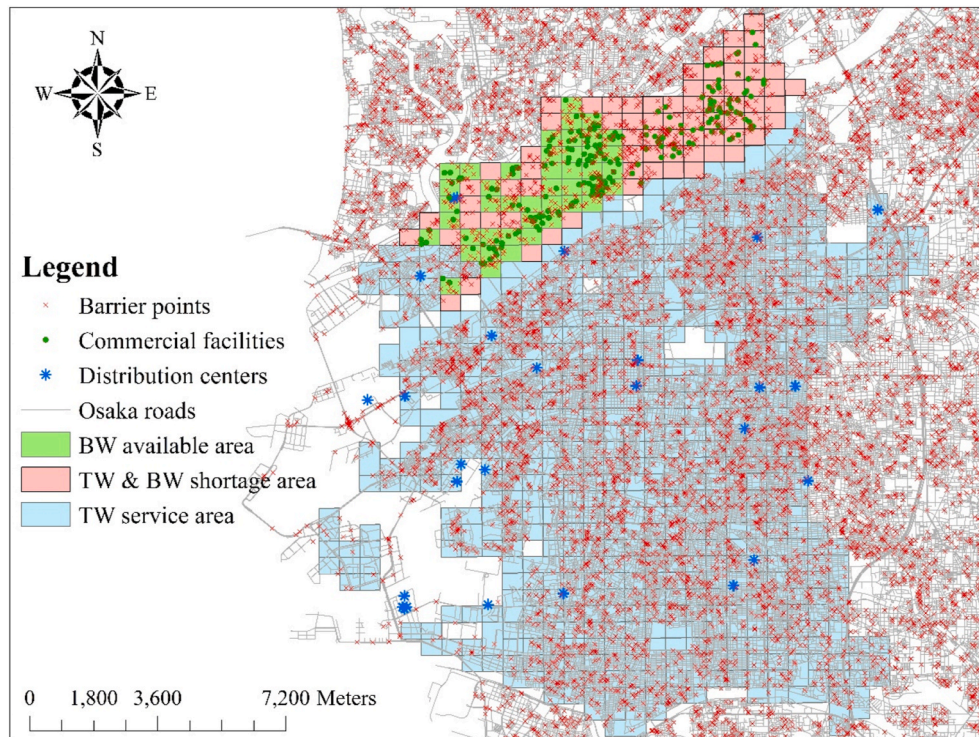


Fig. 6. Availability analysis of bottled water in the disaster scenario.

which would cause a shortage of EW and BW.

#### 4.3.1. Effects of electricity disruption on societal impact

Using the scenario presented in Section 4.2 as a benchmark scenario that had one district suffering from a power outage, we increased the number of districts without electricity sequentially and investigated the

impact (Table 3). Only the societal impact of the reduction in available water due to electricity disruption was investigated, even though electricity disruption can have other negative societal impacts [23]. Under a certain disruption scenario, there were many different combinations of districts without electricity, and only the worst situation is selected to be shown in Table 3. Additionally, the worst combination was considered

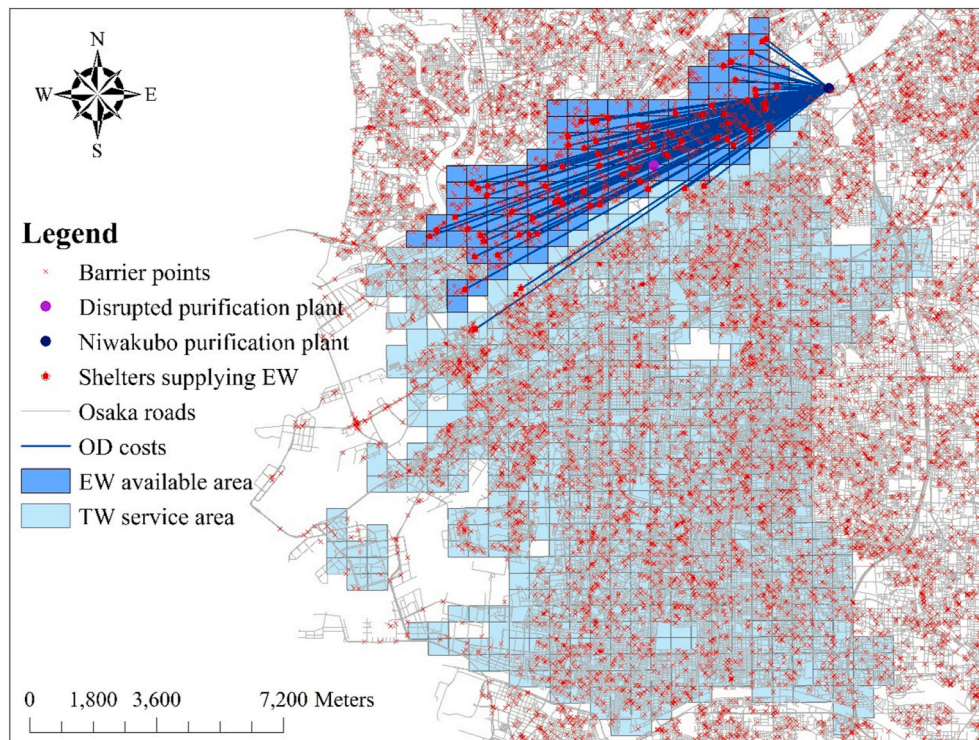


Fig. 7. Availability analysis of emergency water in the disaster scenario.

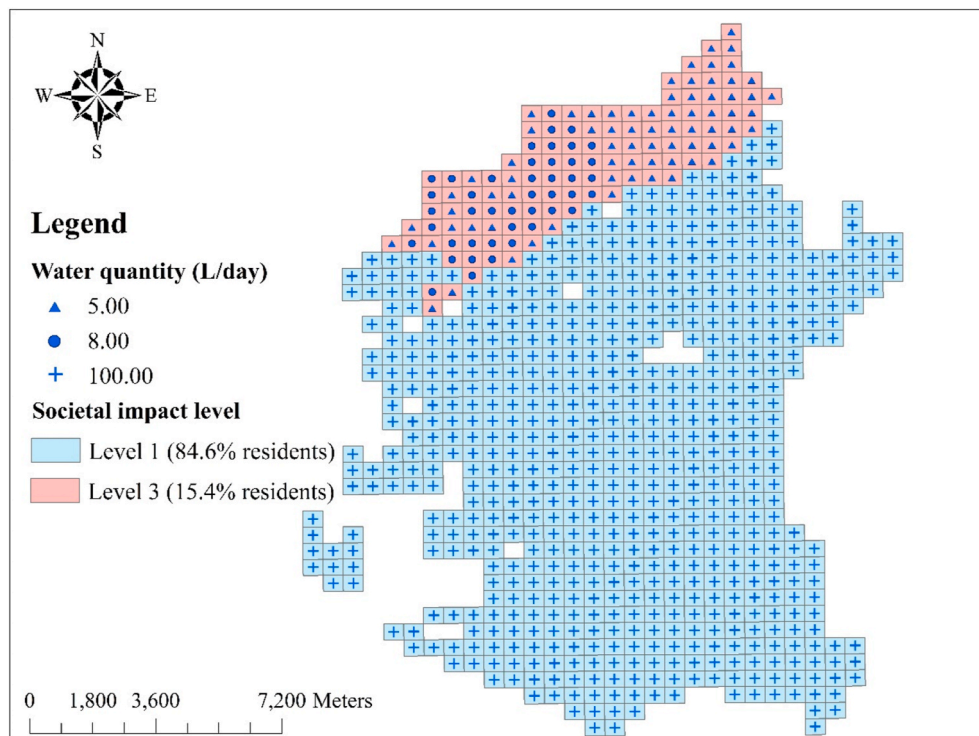


Fig. 8. Spatial distribution of estimated societal impact under the disaster scenario.

as the base scenario (two districts) for the next increasing disruption scenario (three districts). Thus, the importance of each district's electricity supply could be examined.

Overall, the results indicate a high resilience and robust structure of the water supply system in Osaka with regards to electricity disruption (Table 3). For example, there were still over 70% of the population who

can get TW, even though eight districts were without a power supply. The main reason for this high resilience was that TW and sufficient water pressure could be provided by the other two purification plants outside of Osaka, despite the Kunijima purification plant being disrupted.

With the increasing severity of the electricity disruption, the societal impact became larger, which means that an increasing number of people

**Table 3**  
Effects of electricity disruption on societal impact.

| Districts without electricity | Level 1 | Level 2 | Level 3 | Level 4 | Level 5 |
|-------------------------------|---------|---------|---------|---------|---------|
| 1 district                    | 84.59%  | 0%      | 15.41%  | 0%      | 0%      |
| 2 districts                   | 82.87%  | 0%      | 17.13%  | 0%      | 0%      |
| 3 districts                   | 81.81%  | 0%      | 8.81%   | 9.38%   | 0%      |
| 4 districts                   | 80.55%  | 0%      | 10.07%  | 9.38%   | 0%      |
| 5 districts                   | 77.04%  | 0%      | 12.94%  | 10.02%  | 0%      |
| 6 districts                   | 71.91%  | 0%      | 16.05%  | 12.04%  | 0%      |
| 7 districts                   | 71.28%  | 0%      | 16.44%  | 12.28%  | 0%      |
| 8 districts                   | 70.82%  | 0%      | 16.44%  | 12.74%  | 0%      |

could not perform their daily water-related activities. Specifically, the proportion of the population in Level 1 (all needs can be satisfied) had a decreasing trend, and the residents exposed to Level 3 or Level 4 kept increasing. The percentage of residents in Level 3 peaked at 17.13%, and then dropped significantly to 8.81% when there were three districts without electricity. At the same disruption, the people in Level 4 started to appear. This was mainly due to the reduction of EW quantity available to residents (below 5 L/d), which is attributed to the increasing number of shelters requiring EW supply under limited emergency resources. This can also explain why 0% of the population were in Level 2, meaning that the sum of EW and BW was always smaller than 20 L/d with limited emergency resources. Although the quantity of EW decreased as the number of power outage districts increased, the available water for each individual was always larger than 2 L/d; hence, the percentage of people in Level 5 remained at 0%.

Another notable finding was that the proportion of population in Level 1 reduced abruptly when the power-outage districts increased from four to five districts (reduced from 80.55% to 77.04%), and from five to six districts (reduced from 77.04% to 71.91%). To some extent, this characteristic is consistent with catastrophic theory, indicating that the water supply system can tolerate electricity disruptions below a certain threshold (four power-outage districts in Osaka's case), but when the threshold is exceeded, the TW shortage areas increase significantly.

#### 4.3.2. Effects of transportation disruption on societal impact

For the transportation disruption sensitivity analysis, the benchmark was also the illustrative scenario (in Section 4.2), and this section assigned different disruption ratios (1%, 5%, 10%) for road segments in order to check the effect of transportation disruption. For each disruption ratio, transportation network components were disrupted randomly with equal failure probability. Additionally, to avoid the uncertainty of random disruption, 100 cases were simulated for each disruption ratio to explore the average and worst societal impact (shown in Table 4).

The results indicate that the average impact increased slightly with increasing disruption ratio. This was mainly reflected in the increasing percentage of residents in Level 4 (0.01% to 1.2%) and Level 5 (0.01% to 0.36%). The percentage of residents in Level 1 remained unchanged (84.59%) because transportation disruption did not affect the TW supply. With increasing disruption ratio, the percentage of residents in Level 3 gradually declined, and more people's survival needs could not be

**Table 4**  
Estimated societal impact under 1%, 5%, 10% road segment disruption scenarios.

| Scenarios      |            | Level 1 | Level 2 | Level 3 | Level 4 | Level 5 |
|----------------|------------|---------|---------|---------|---------|---------|
| 1% disruption  | Average    | 84.59%  | 0.00%   | 15.39%  | 0.01%   | 0.01%   |
|                | Worst case | 84.59%  | 0.00%   | 15.16%  | 0%      | 0.26%   |
| 5% disruption  | Average    | 84.59%  | 0.00%   | 15.26%  | 0.06%   | 0.09%   |
|                | Worst case | 84.59%  | 0.00%   | 14.65%  | 0.4%    | 0.37%   |
| 10% disruption | Average    | 84.59%  | 0.00%   | 13.84%  | 1.20%   | 0.36%   |
|                | Worst case | 84.59%  | 0.00%   | 6.36%   | 8.43%   | 0.62%   |

satisfied. The reason for this phenomenon was the reduction in EW and BW caused by increasing travel costs and the inaccessibility of emergency shelters or commercial facilities.

Comparing the average results with the worst-case results, we found that the uncertainty of societal impact increased with the higher disruption ratios. As shown in Table 4, when the disruption ratio was 10%, there were significant differences between the average societal impact (13.84% in Level 3 and 1.20% in Level 4) and the worst case societal impact (6.36% in Level 3 and 8.43% in Level 4). In this typical case study, the societal impact of transportation disruption was not significant; however, it is still possible that a high societal impact could occur under a higher disruption ratio. This result, coupled with the study by Mostafavi et al. [33], emphasises the importance of the transportation system in water supply resilience.

## 5. Discussion and conclusion

In a disaster event, disruptions to CI directly reduce the supply of life-supporting resources/services, which leads to the unsatisfaction of residents' physiological needs and further causes societal impact. However, societal impact cannot be easily quantified due to the complexity of connecting it with infrastructure disruptions. The dependency and adaptive capacity of CI systems in a disaster event increase the difficulties in accurately estimating negative effects on society. To fill this gap, this study focused on the water analysis, and proposes a theoretical framework and integrated methodology to estimate the corresponding societal impact due to related infrastructure disruptions. The integrated methodology combines social and engineering aspects. From the analysis of social dimensions, a quantitative relationship between needs satisfaction level and available water quantity is established. As for engineering analysis, the available water quantity for individuals under disruptions of infrastructure is estimated using the integrated water quantity estimation model. This model includes availability analyses of TW, BW, and EW based on the hydraulic characteristics of the water network, failure causes of the commercial facilities, and the equal distribution principle of emergency services, respectively. As such, the severity and spatial distribution of the societal impact caused by a disaster can be calculated.

The illustrative case study using data from the city of Osaka demonstrated that the spatial distribution of societal impact can be derived from the proposed methodology, which can provide essential information for emergency management. The results indicate that the impact was dominated by the availability of TW, followed by EW and BW under this specific scenario. Additionally, the results highlight the importance of the adaptive capacity of water supply infrastructure and are consistent with previous work [33]. Specifically, 15.41% of the residents of Osaka were in Level 3 (survival needs can be met) for our illustrative case study, but these people would have been in Level 5 (no needs can be met) if TW and BW were not included. However, in practice, several countries have developed emergency response strategies to deliver EW to water-shortage areas. For example, the distribution of EW in Osaka city of Japan is based on the equality principle, and other policies can be examined according to the background of different cities.

Sensitivity analyses were conducted to investigate the effects of disrupted CI on society and to validate the proposed model. The societal impact became larger with increasing severity of power and transportation disruptions. Specifically, when power outage areas increase from one to eight districts, the percentage of residents in Level 4 or Level 5 increases from 0% to 12.74%. As for transportation disruption, in the worst case, the percentage of residents in Level 4 or Level 5 increase from 0.26% to 9.05% when the disruption ratio increased from 1% to 10%. Furthermore, compared with road segment failure (15.41% of residents were not in Level 1), power outages affected more people (29.18% of residents were not in Level 1); because the availability of TW could be influenced by the disrupted electricity instead of transportation disruption. However, it is still possible to have a large impact (8.43% of

people are in Level 4) when shelters or commercial facilities become isolated due to severe transportation disruption (10% road segment disruption). In summary, to avoid a large societal impact, the number of power outage districts needs to be kept below five in the Osaka case study, and inaccessible regions caused by transportation disruption should be emphasised.

The proposed framework and methodology can provide valuable empirical information for city managers regarding disaster prevention/mitigation and emergency rescue, allowing them to maximise the possibility of maintaining residents' normal activities and reduce the negative societal impact. Although this study validates the proposed methodology using a water analysis in a disaster scenario, the framework can also be applied to other essential goods/services analyses. Future work could extend our proposed model by considering the evacuation of residents after disasters, because regional population redistribution would affect the availability of TW and EW. The post-disaster recovery of different infrastructure systems could also be included in this model, such that the dynamic societal impact and recovery strategies in a city can be explored. Analysis on other basic individual survival needs (e.g. food, housing, and medical services) under disruptions should be integrated into the societal impact evaluation. To predict the societal impact more accurately, failure probability of infrastructure components and different emergency policies should also be incorporated and simulated in future studies.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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